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BASED IN Genoa, Italy

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SECTOR RTD-A · IINF-01/A

UPDATED In role since March 2025

01 Profile

Assistant Professor (RTD-A) at DITEN, University of Genoa, in the Electronics sector (IINF-01/A), since March 2025. PhD in Science and Technology for Electronic and Telecommunication Engineering, 2024.

Research on embedded machine learning and edge AI: TinyML toolchains, quantized and binary neural networks on microcontrollers and FPGAs, perception and decision-making for automated driving, deep reinforcement learning, and on-device voice assistants. The work is carried out within European and national projects with automotive and industrial partners.

Edge AI

TinyML

Embedded deep learning

Automated driving

Deep reinforcement learning

Embedded voice assistants

02 Appointments & education

POSITIONS

Mar 2025 – present **Assistant Professor (RTD-A)**

DITEN, University of Genoa

Scientific sector IINF-01/A — Electronics.

Nov 2023 – Feb 2025 **Research Fellow**

DITEN, University of Genoa

Grant: “Machine Learning models for the processing of data from autonomous vehicles”.

Nov 2020 – Oct 2023 **PhD Candidate**

DITEN, University of Genoa — XXXVI cycle

Scholarship: “Deep learning on the edge”.

EDUCATION

2024

PhD in Science and Technology for Electronic and Telecommunication Engineering

University of Genoa — awarded 14 March 2024

Thesis: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”

2020	MSc in Electronic Engineering (LM-29) University of Genoa — 24 July 2020 Thesis: “Design and implementation of a multiplatform execution engine for IoT applications” · Grade: 110/110
2017	BSc in Electronic Engineering and Information Technologies (L-8) University of Genoa — 27 October 2017 Thesis: “Model-based design of Lego robot control in Simulink environment”
2014	Scientific High School Diploma Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV) Grade: 100/100

Chartered Information Engineer — State Examination passed in the second session of 2020.

03 Bibliometrics

SYNCED 17 SEPT 2026

66 SCOPUS DOCUMENTS	336 CITATIONS	11 H - INDEX	71 CO - AUTHORS
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Every figure here comes from Scopus (author ID 57220892898) and is refreshed automatically. The 6 national-conference and workshop items that Scopus does not index are listed below, but contribute to none of them.

04 Publications

72 ITEMS · 66 IN SCOPUS

2026	Cross-platform deployment and characterization of an on-device generative conversational agent P. Firpo, R. Berta, F. Bellotti, L. Lazzaroni , M. Fresta, A. Pighetti, A. Abedinia, N. Yoonesi, D.P. Pau <i>Journal of Systems Architecture</i> vol. 181, art. 103981 Journal https://doi.org/10.1016/j.sysarc.2026.103981
	A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction L. Forneris, R. Berta, M. Fresta, L. Lazzaroni , H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti <i>IEEE Signal Processing Letters</i> vol. 33, pp. 136–140 Journal https://doi.org/10.1109/lsp.2025.3638676
	FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction A. Saad, R. Berta, M. Fresta, L. Lazzaroni , A. Pighetti, O. Srour, F. Tango, F. Bellotti <i>IEEE Transactions on Circuits and Systems II: Express Briefs</i> vol. 73, no. 8, pp. 888–892 Journal https://doi.org/10.1109/tcsii.2026.3705103
	Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms H. Rojhan, F. Bellotti, M. Fresta, L. Lazzaroni , R. Berta <i>IEEE Access</i> vol. 14, pp. 44886–44903 Journal https://doi.org/10.1109/access.2026.3676567
	StabilEdge, Formal Verification of Edge AI Applications L. Lazzaroni , R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti <i>IEEE Access</i> vol. 14, pp. 37416–37434 Journal https://doi.org/10.1109/access.2026.3667991

Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers

A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 321–327

Conference

https://doi.org/10.1007/978-3-032-17174-0_46

Preface

M. Ruó Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero

Lecture Notes in Electrical Engineering pp. v–vi

Editorial

Optimizing Binary AutoEncoders on STM32 Microcontrollers

H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 301–307

Conference

https://doi.org/10.1007/978-3-032-17174-0_43

Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students

L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 242–248

Conference

https://doi.org/10.1007/978-3-032-17174-0_34

Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform

H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 218–223

Conference

https://doi.org/10.1007/978-3-032-17174-0_30

Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA

D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti

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https://doi.org/10.1007/978-3-032-17174-0_4

Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning

O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 212–217

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https://doi.org/10.1007/978-3-032-17174-0_29

Implementing YoloV8 on Zynq UltraScale + MPSoC FPGA Boards

H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 289–294

National

https://doi.org/10.1007/978-3-032-17305-8_31

FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving

A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 295–301

National

https://doi.org/10.1007/978-3-032-17305-8_32

Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision

A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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https://doi.org/10.1007/978-3-032-17174-0_12

Embedded Deployment of an LLM-Based Voice Assistant

L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srou, R. Berta

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https://doi.org/10.1007/978-3-032-17174-0_39

Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection

V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**

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A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors

S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti

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A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems

E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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https://doi.org/10.1007/978-3-032-17174-0_35

A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents

A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti

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2025

An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection

L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta

IEEE Sensors Journal vol. 25, no. 24, pp. 44859–44874

Journal

<https://doi.org/10.1109/jsen.2025.3627477>

TinyML Acceleration with MAX78000

A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta

Proceedings of SIE 2024 Lecture Notes in Electrical Engineering pp. 468–474

National

https://doi.org/10.1007/978-3-031-71518-1_57

Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection

L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**

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https://doi.org/10.1007/978-3-031-84100-2_19

Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking

H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti

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https://doi.org/10.1007/978-3-031-84100-2_46

Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking

F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**

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https://doi.org/10.1007/978-3-031-84100-2_17

RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset

R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti

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https://doi.org/10.1007/978-3-031-84100-2_47

RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

IEEE Sensors Letters vol. 9, no. 9, art. 6010204

Journal

<https://doi.org/10.1109/lensens.2025.3600982>

Quantization of MobileNetV2 for Resource-Constrained Microcontrollers

L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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Conference

https://doi.org/10.1007/978-3-031-84100-2_59

Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board

H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

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https://doi.org/10.1007/978-3-031-84100-2_29

On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection

M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta

IEEE Open Journal of the Industrial Electronics Society vol. 6, pp. 1351–1369

Journal

<https://doi.org/10.1109/ojies.2025.3601982>

Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices

A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti

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pp. 323–330

Conference

https://doi.org/10.1007/978-3-031-84100-2_38

Green Machine Learning for the IoT (GEMINI): A Position Paper

L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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https://doi.org/10.1007/978-3-031-84100-2_40

Exploring Decision Transformer for Highway Automated Driving

L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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https://doi.org/10.1007/978-3-031-84100-2_15

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta

IEEE Access vol. 13, pp. 26589–26607

Journal

<https://doi.org/10.1109/access.2025.3539392>

Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection

M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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https://doi.org/10.1007/978-3-031-84100-2_30

Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices

A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti

2025 IEEE International Conference on Consumer Electronics (ICCE)

Digest of Technical Papers IEEE International Conference on Consumer Electronics

Conference

<https://doi.org/10.1109/icce63647.2025.10930047>

Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification

A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

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https://doi.org/10.1007/978-3-031-84100-2_28

2024

An embedded end-to-end voice assistant

L. Lazzaroni, F. Bellotti, R. Berta

Engineering Applications of Artificial Intelligence vol. 136, art. 108998

Journal

<https://doi.org/10.1016/j.engappai.2024.108998>

Adversarial Policy Generation in Automated Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving

Workshop

CBin-NN: An Inference Engine for Binarized Neural Networks

F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti

Electronics vol. 13, no. 9, art. 1624

Journal

<https://doi.org/10.3390/electronics13091624>

End-to-End Dataset Collection System for Sport Activities

M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta

Electronics vol. 13, no. 7, art. 1286

Journal

<https://doi.org/10.3390/electronics13071286>

YoloP-Based Pre-processing for Driving Scenario Detection

M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti

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https://doi.org/10.1007/978-3-031-48121-5_60

Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection

M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta

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https://doi.org/10.1007/978-3-031-48121-5_64

Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations

A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta

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https://doi.org/10.1007/978-3-031-48121-5_18

Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity

R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti

Journal of Intelligent and Connected Vehicles vol. 7, no. 3, pp. 229–244

Journal

<https://doi.org/10.26599/jicv.2023.9210039>

Developing a TinyML Image Classifier in an Hour

R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 946–960

Journal

<https://doi.org/10.1109/ojies.2024.3451959>

Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving

L. Lazzaroni, A. Pighetti, F. Bellotti, R. Berta

Proceedings of SIE 2023 Lecture Notes in Electrical Engineering pp. 351–356

National

https://doi.org/10.1007/978-3-031-48711-8_42

Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge

A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 781–794

Journal

<https://doi.org/10.1109/ojies.2024.3434341>

Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach

L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta

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Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents

F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion

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pp. 170–180

Conference

https://doi.org/10.1007/978-3-031-49065-1_17

2023

Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles

L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta

International Journal of Serious Games vol. 10, no. 4, pp. 153–170

Journal

<https://doi.org/10.17083/ijsg.v10i4.638>

Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild

F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion

Applied Sciences vol. 13, no. 11, art. 6432

Journal

<https://doi.org/10.3390/app13116432>

Supporting a .csv-based Workflow in MongoDB for Data Analysts

M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta

2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)

IEEE International Symposium on Industrial Electronics

Conference

<https://doi.org/10.1109/isie51358.2023.10228044>

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

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https://doi.org/10.1007/978-3-031-30333-3_41

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

IEEE Access vol. 11, pp. 28522–28550

Journal

<https://doi.org/10.1109/access.2023.3259544>

Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System

M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta

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https://doi.org/10.1007/978-3-031-30333-3_2

Developing a Toolchain for Synthetic Driving Scenario Datasets

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

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https://doi.org/10.1007/978-3-031-30333-3_29

Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

Proceedings of SIE 2022 Lecture Notes in Electrical Engineering pp. 239–244

National

https://doi.org/10.1007/978-3-031-26066-7_37

Deep Reinforcement Learning for Automated Car Parking

L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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2022

Synthetic Driving Scenarios Generator

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

National

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

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Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**

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An Embedded End-to-End Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

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Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications

R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**

Sensors vol. 22, no. 3, art. 713

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<https://doi.org/10.3390/s22030713>

High-Level Decision-Making Non-player Vehicles

A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

Games and Learning Alliance

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https://doi.org/10.1007/978-3-031-22124-8_22

Developing a Synthetic Dataset for Driving Scenarios

J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora

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https://doi.org/10.1007/978-3-030-95498-7_43

Classifying Simulated Driving Scenarios from Automated Cars

M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta

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https://doi.org/10.1007/978-3-030-95498-7_32

A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses

F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta

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Project and Development of an Italian Offline Embedded Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste

National

Edgine, A Runtime System for IoT Edge Applications

R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**

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https://doi.org/10.1007/978-3-030-66729-0_31

Adapting Autonomous Agents for Automotive Driving Games

G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 101–110

Conference

https://doi.org/10.1007/978-3-030-92182-8_10

2020

Employing an IoT Framework as a Generic Serious Games Analytics Engine

L. Lazzaroni, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 79–88

Conference

https://doi.org/10.1007/978-3-030-63464-3_8

05 Research projects

Synergies

Generation of scenarios for development, training, virtual testing and validation of CCAM systems

European — HORIZON-CL5-2023-D6-01-02 · Member of the University of Genoa research unit

Generates test scenarios for training, testing and validating CCAM systems, focusing on urban and rural traffic — where relevant-scenario knowledge is far thinner than for highway driving. Runs with most major European carmakers and universities active in the field.

Hi-Drive

Addressing challenges toward the deployment of higher automation

European — H2020, GA 101006664 · Member of the University of Genoa research unit

Develops and tests advanced automated driving at scale, moving from conditional automation (SAE L3) to high automation across different scenarios and road contexts.

GEMINI

Green Machine Learning for the IoT

National — PRIN 2022 · Member of the University of Genoa research unit (DII–UNITN / DITEN–UNIGE)

A framework for fully sustainable TinyML applications deployed along an edge-to-cloud continuum, targeting battery-free embedded devices — including those with no network connectivity at all.

ADINA

Integrated Digital Assistant Automotive

Regional — POR FESR Piedmont 2014-2020 · Member of the University of Genoa research unit, with Lifetouch

An in-vehicle virtual assistant on dedicated hardware, connected to the car's systems: AI algorithms interpret the driver's requests and act on vehicle and infotainment functions.

06 Teaching & supervision

A.Y. 2024/25, 2025/26 **Distributed Electronic Systems and Technologies for Environmental Monitoring**

Course lecturer · cod. 94631

A.Y. 2024/25, 2025/26	Tiny Machine Learning Course lecturer · cod. 114752
since A.Y. 2022/23	Machine Learning for Automated Driving Subject expert · cod. 108716
A.Y. 2023/24	Computer Graphics Arch. — Games and Simulation Adjunct professor · cod. 172317
A.Y. 2022/23	Autonomous Agents in Games Teaching support · cod. 198216
A.Y. 2020/21	Fundamentals of Programming of Electronic Systems Teaching support · cod. 166061

All courses at the University of Genoa. Innovative teaching methodologies were also introduced in “Fundamentals of Programming of Electronic Systems” (166061) and “Makers Approach to Electronic Design” (186985), A.Y. 2020/21.

SUPERVISION

12	BSc theses — Electronic Engineering and Information Technologies (L-8)	Co-supervisor of Ali Zare, PhD Programme in Science and Technology for Electronic and Telecommunication Engineering (XLI cycle).
10	MSc theses — Electronic Engineering (LM-29)	
7	MSc theses — Technological Engineering for Strategy and Security (LM/DS)	Member of the Teaching and Admissions Committee, MSc in Engineering for Natural Risk Management.
2	MSc theses — Computer Science (LM-18)	

07 Editorial & service

EDITORIAL ROLES

2025	Editorial Board Member (Early Career) Sensors — MDPI Since 18 December 2025.
2025	Volume Editor APPLEPIES 2025 — Springer LNEE Proceedings of the International Conference on Applications in Electronics Pervading Industry, Environment and Society.
2024	Guest Editor Energies — MDPI Special Issue “Tiny Machine Learning for Energy Applications”, April 2024 — August 2025.

CONFERENCE ORGANISATION

Program Chair APPLEPIES 2025 Turin, Italy, 11–12 September 2025.
Program Committee APPLEPIES Since 2023.
Program Committee GALA — Games and Learning Alliance Since 2023.
Session Chair APPLEPIES 2023 1st Short Oral session — Genoa, Italy, 28–29 September 2023.

PEER REVIEW

JOURNALS	CONFERENCES
IEEE Access since 2023	IEEE ICRA — International Conference on Robotics and Automation since 2026
Journal of Systems Architecture since 2023	IEEE/RSJ IROS — Intelligent Robots and Systems since 2025
Engineering Applications of Artificial Intelligence since 2024	GALA — Games and Learning Alliance since 2023, 2024
Sensors since 2024	APPLEPIES since 2021–2024
IET Intelligent Transport Systems since 2025	
IEEE Transactions on Intelligent Vehicles since 2026	
IEEE Transactions on Circuits and Systems II: Express Briefs since 2026	
Robotics and Autonomous Systems since 2026	
Scientific Reports since 2026	
Journal of Big Data since 2026	
Discover Internet of Things since 2026	

11–14 Jan 2025	43rd IEEE ICCE – International Conference on Consumer Electronics Las Vegas, NV, USA
19–20 Sep 2024	APPLEPIES 2024 Turin, Italy
24–26 Jun 2024	55th Annual Meeting of the Italian Society of Electronics (SIE) Genoa, Italy
28–29 Sep 2023	APPLEPIES 2023 Genoa, Italy
7–9 Sep 2023	54th Annual Meeting of the Italian Society of Electronics (SIE) Noto (RG), Italy
26–27 Sep 2022	APPLEPIES 2022 Genoa, Italy
7–9 Sep 2022	53rd Annual Meeting of the Italian Society of Electronics (SIE) Pizzo (VV), Italy
21–22 Sep 2021	APPLEPIES 2021 Pisa, Italy
1–2 Dec 2021	GALA 2021 – Games and Learning Alliance La Spezia, Italy (online)
7–9 Jul 2021	52nd Annual Meeting of the Italian Society of Electronics (SIE) Trieste, Italy (online)
9–10 Dec 2020	GALA 2020 – Games and Learning Alliance Laval, France (online)
19–20 Nov 2020	APPLEPIES 2020 Online

09 Awards & memberships

2025	Best Presentation of the Session – “Edge Computing” 43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 January 2025 A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti – “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”
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PROFESSIONAL MEMBERSHIPS

IEEE

IEEE Member.

IEEE Consumer Technology Society

Member of the “Machine Learning, Deep Learning and AI in CE (MDA)” Technical Committee.

SIE — Italian Society of Electronics

Member within the University of Genoa Research Unit since 2021, research area “Electronic systems and applications”.

10 Contact & profiles

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LINKEDIN luca-lazzaroni		

Publications and indicators are pulled from Scopus by a scheduled job, and the PDF is printed from this page — the two cannot drift apart.

Full CV as a PDF. Source of this site on GitHub. Set in IBM Plex.